



Summer Term 2026

Investment and Financial Management
Exercise on Mathematical Basics

Session 05: Redemptions

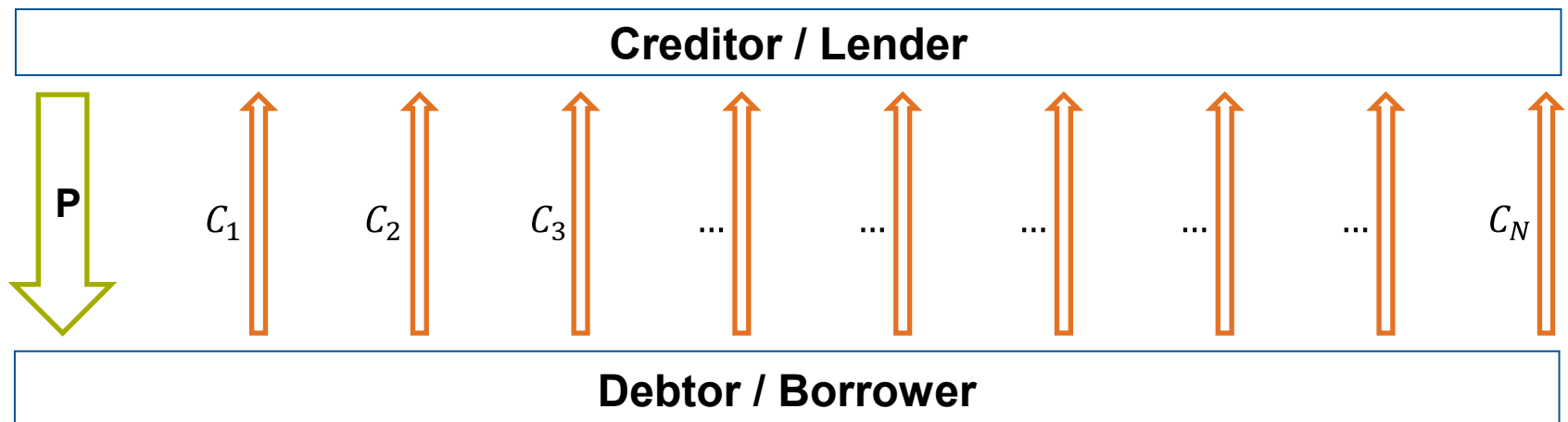
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- 3.2 Instalment repayment**
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Redemption or repayment calculation is about the repayment of an outstanding principal loan amount from the borrower to the lender



The provided loan of the principal loan amount P can be repaid by a series of regular periodic annuity-immediate payments C_k with k being the corresponding period. All of these payments consist of a **repayment amount** T_k and a **periodic interest payment** I_k .

Important formula

k – period

r – interest rate

A_k – Annuity payment in period k

T_k – Repayment amount in period k

I_k – Interest payment in period k

D_k – Remaining loan amount in period k

D_0 – Principal, i. e. loan amount in period 0

$$(1): A_k = T_k + I_k$$

$$(2): D_k = D_{k-1} - T_k$$

$$(3): D_0 = \sum_{k=1}^N T_k$$

$$(4): I_k = r * D_{k-1}$$

$$(5): D_0 = \sum_{k=1}^N A_k * (1 + r)^{-k}$$

Derivation of the repayment formula: $D_N = f(D_0)$

For period $t = 1$:

$$(1): T_1 = A_1 - r * D_0 \Rightarrow D_1 = D_0 - T_1 = D_0 - (A_1 - r * D_0) = (1 + r) * D_0 - A_1$$

For period $t = 2$:

$$(2): I_2 = r * D_1 = r * ((1 + r) * D_0 - A_1)$$

$$(3): T_2 = A_2 - r * (1 + r) * D_0 + r * A_1$$

$$(4): D_2 = D_1 - T_2 = (1 + r) * D_0 - A_1 - A_2 + r * (1 + r) * D_0 - r * A_1$$

$$(5): D_2 = (1 + r)^2 * D_0 - (1 + r) * A_1 - A_2$$

$$(6): D_N = (1 + r)^N * D_0 - \sum_{k=1}^N (1 + r)^{N-k} * A_k$$

$$(7): \text{Exception: } D_{\text{Final period}} = 0$$

$$(8): D_0 = \sum_{k=1}^N (1 + r)^{-k} * A_k$$

Loan repayment plan

Period k	Loan amount (period beginning) $SD_k = D_{k-1}$	Interest payment I_k	Loan repayment T_k	Annuity payment A_k	Remaining loan amount D_k
1	D_0	$D_0 * r$	T_1	$T_1 + I_1$	$D_0 - T_1$
2	$D_0 - T_1$	$(D_0 - T_1) * r$	T_2	$T_2 + I_2$	$D_0 - T_2 - T_1$
...	...	...	...	...	...
...	...	...	...	...	...
...	...	...	...	...	...
N	T_N	$T_N * r$	T_N	$T_N + I_N$	0
$N + 1$	0				

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If the repayment amounts T_k are equal over the whole redemption period, the repayments are called instalment repayments.

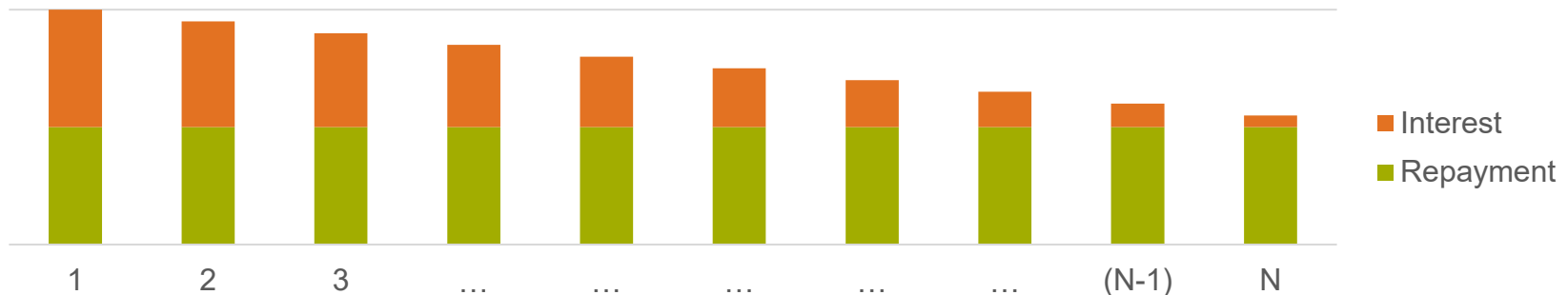
- Idea:** No variation in the repayment amount: $T_1 = T_2 = T_3 = \dots = T_N = T$

$$(1): T_k = T = \frac{D_0}{N}$$

$$(2): I_k = D_{k-1} * r$$

$$(3): D_{k-1} = D_0 * \left(1 - \frac{k-1}{N}\right) = T * (N - (k - 1)) = \frac{D_0}{N} * (N - (k - 1))$$

$$(4): A_k = \frac{D_0}{N} + D_{k-1} * r = \frac{D_0}{N} + \frac{D_0}{N} * (N - k + 1) * r = \frac{D_0}{N} * [1 + (N - k + 1) * r]$$



(A.1) A loan of 36,000 EUR should be repaid by instalment payments within three years. Please assume an annual interest rate of 10% and draw the repayment plan.

k	$SD_k = D_{k-1}$	I_k	T_k	A_k	D_k
1	36,000	3,600	12,000	15,600	24,000
2	24,000	2,400	12,000	14,400	12,000
3	12,000	1,200	12,000	13,200	0
$N + 1$	0				

$$(1): T = \frac{D_0}{N} = \frac{36,000}{3} = 12,000 \text{ EUR}$$

$$(2): I_k = D_{k-1} * r \Rightarrow I_1 = 36,000 * 0.10 = 3,600 \text{ EUR}$$

$$(3): A_k = T_k + I_k \Rightarrow A_1 = 12,000 + 3,600 = 15,600 \text{ EUR}$$

$$(4): D_k = D_{k-1} - T \Rightarrow D_1 = 36,000 - 12,000 = 24,000 \text{ EUR}$$

Annuity payments follow an arithmetic series

★ (A.2) A loan of 50,000 EUR is repaid by an annuity of 20 equal instalment payments at the end of each year. The interest rate is considered to be stable at 6% which needs to be paid together with the annual instalment payment.

(a) How much interest need to be paid in the 10th, 15th, and in the last year?

$A = 600 \text{ EUR},$
 $300 \text{ EUR}, 100 \text{ EUR}$

$B = 150 \text{ EUR},$
 $150 \text{ EUR}, 150 \text{ EUR}$

$C = 1,650 \text{ EUR},$
 $900 \text{ EUR}, 150 \text{ EUR},$

$D = 900 \text{ EUR},$
 $900 \text{ EUR}, 900 \text{ EUR}$

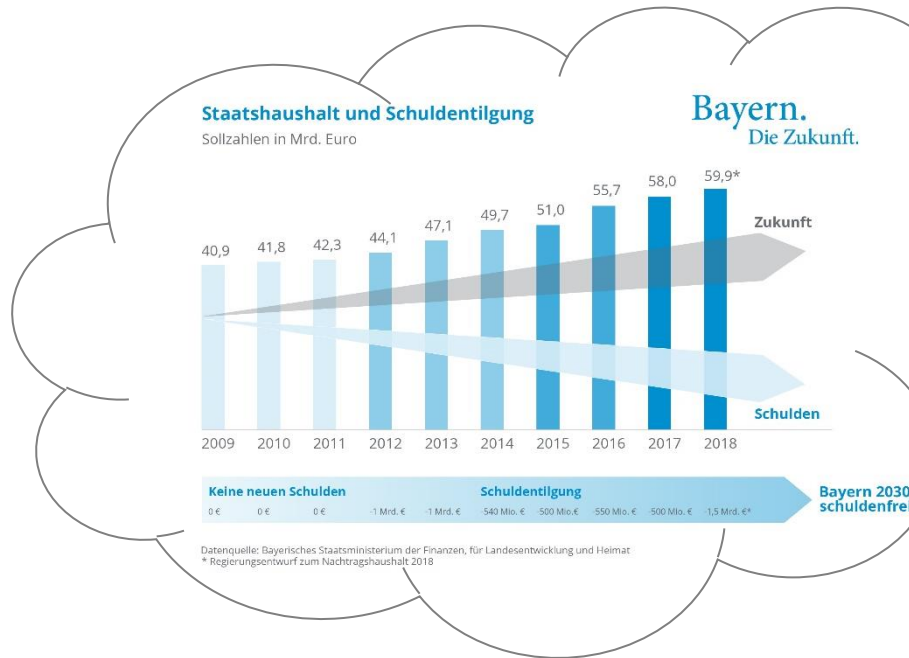
$$(1): I_k = D_{k-1} * r = D_0 * \left(1 - \frac{(k-1)}{N}\right) * r$$

$$(2): I_{10} = 50,000 \text{ EUR} * \left(1 - \frac{(10-1)}{20}\right) * 0.06 = 1,650 \text{ EUR}$$

$$(3): I_{15} = 900 \text{ EUR}$$

$$(4): I_{20} = 150 \text{ EUR}$$

Loan Redemption: Example



**No more debt
in 2030!**



Period k	Loan amount (period beginning) $SD_k = D_{k-1}$	Interest payment I_k	Loan repayment T_k	Annuity payment A_k	Remaining loan amount D_k
1	D_0	$D_0 * r$	T_1	$T_1 + I_1$	$D_0 - T_1$
2	$D_0 - T_1$	$(D_0 - T_1) * r$	T_2	$T_2 + I_2$	$D_0 - T_2 - T_1$
...	...	...	...	...	...
...	...	...	...	...	...
...	...	...	...	...	...
N	T_N	$T_N * r$	T_N	$T_N + I_N$	0
$N + 1$	0				

Instalment repayment: Example

k	$SD_k = D_{k-1}$	$I_k (r = 2.58\%)$	T_k	A_k	D_k
2017	29.000	0.747	2.0714	2.820	26.929
2018	26.929	0.695	2.0714	2.766	24.857
2019	24.857	0.641	2.0714	2.713	22.786
2020	22.785	0.588	2.0714	2.659	20.714
2021	20.714	0.534	2.0714	2.606	18.643
2022	18.643	0.481	2.0714	2.552	16.572
2023	16.572	0.428	2.0714	2.499	14.500
2024	14.500	0.374	2.0714	2.446	12.429
2025	12.429	0.321	2.0714	2.392	10.357
2026	10.357	0.267	2.0714	2.339	8.286
2027	8.286	0.214	2.0714	2.285	6.215
2028	6.215	0.160	2.0714	2.232	4.143
2029	4.143	0.107	2.0714	2.178	2.072
2030	2.072	0.053	2.0714	2.125	0

Note: Values for 2017 are given (Bayerische Staatszeitung (2017)), Values in billion EUR.

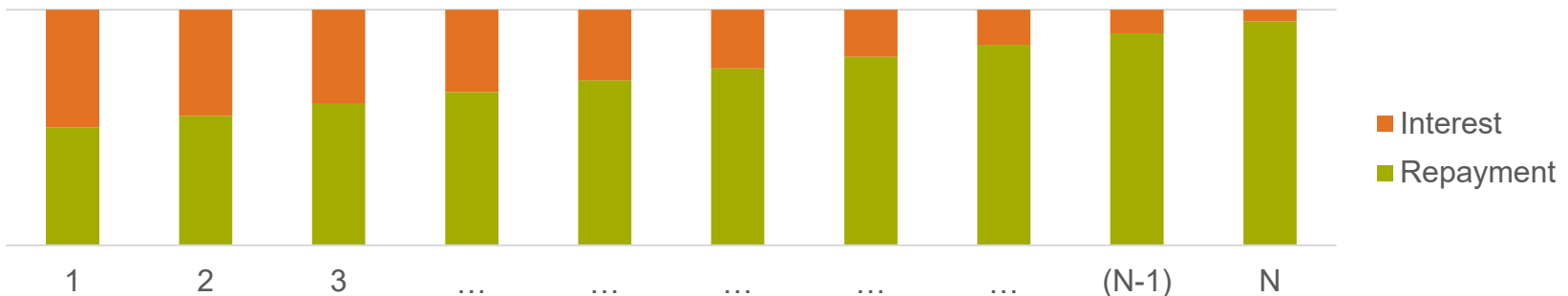
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If the annuity amounts A_k are equal over the whole redemption period, the repayments are called annuity repayments.

- **Idea:** In every annuity period the annuity payment remains equal. Rather than the repayment amount, now the annuity amount is constant.
- (1): $A_1 = A_2 = A_3 = \dots = A_N = A$
- Determination as a constant annuity-immediate with a present value of D_0 of the annuity A and a duration of N : Make use of annuity calculation!

(2): $A = D_0 * \frac{q^N * (q - 1)}{q^N - 1}$

Present Value Formula for an Annuity-immediate



If the annuity amounts A_k are equal over the whole redemption period, the repayments are called annuity repayments.

The remaining loan amount D_{k-1} equals the sum of the initial loan amount and the interest which need to be paid until period k ($D_0 * q^{k-1}$) minus the current time value at k of the sum of all repayments made until period k ($A * \frac{q^{k-1}-1}{q-1}$).

$$(1): D_{k-1} = D_0 * q^{k-1} - A * \frac{q^{k-1} - 1}{q - 1} = D_0 * q^{k-1} - D_0 * \frac{q^N}{q^N - 1} * (q^{k-1} - 1)$$

$$(2): D_{k-1} = D_0 * \left(q^{k-1} - \frac{q^N * (q^{k-1} - 1)}{q^N - 1} \right) = D_0 * \frac{-q^{k-1} + q^N}{q^N - 1}$$

$$(3): D_{k-1} = D_0 * \frac{q^N - q^{k-1}}{q^N - 1}$$

For annuity repayments, it holds: $A_k = A_{k-1}$

$$(1): I_k + T_k = I_{k-1} + T_{k-1} \Leftrightarrow T_k = T_{k-1} + I_{k-1} - I_k$$

$$(2): T_k = T_{k-1} + r * \underbrace{(D_{k-2} - D_{k-1})}_{\substack{T_{k-1} \\ I_{k-1} - I_k}} = (1 + r) * T_{k-1} = q * T_{k-1} \underbrace{=}_{T_{k-1} = q * T_{k-2}} q * (q * T_{k-2}) = \dots$$

We can apply the concept of a geometric series here.

$$(3): T_k = q^{k-1} * T_1 \quad \text{with } T_1 = \underbrace{A}_{D_0 = A \dots} - \underbrace{r * D_0}_{I_1} \underbrace{=}_{\frac{r}{q^N - 1} * D_0}$$

$$A = D_0 * \frac{q^N * (q - 1)}{q^N - 1}$$

$$(4): T_k = r * \frac{q^{k-1}}{q^N - 1} * \underbrace{D_0}_{D_0 = A \dots} \underbrace{=}_{A} A * \frac{1}{q^{N-k+1}}$$

How to solve for the interest payment?

$$(1): I_k = A - T_k = r * D_{k-1}$$

$$(2): I_k = A - A * \frac{1}{q^{N-k+1}} = A * \left(1 - \frac{1}{q^{N-k+1}} \right)$$

$$T_k = A * \frac{1}{q^{N-k+1}}$$

$$(3): I_k = r * D_{k-1} = D_0 * \frac{r * (q^N - q^{k-1})}{q^N - 1}$$

$$D_{k-1} = D_0 * \frac{q^N - q^{k-1}}{q^N - 1}$$

(A.3) Due to the renovation of his house the house owner needs to take on a loan with an amount of 36,000 EUR which should be repaid within the next three years. The interest rate equals 10% and the annuity payments should be made at the end of each year. Please draw a repayment plan.

$$(1): A = D_0 * \frac{q^N * (q - 1)}{q^N - 1} = 36,000 * \frac{1.1^3 * (1.1 - 1)}{1.1^3 - 1} = 14,476.13 \text{ EUR}$$

k	$SD_k = D_{k-1}$	I_k	T_k	A_k	D_k
1	36,000.00	3,600	10,876.13	14.476.13	25,123.87
2	25,123.87	2,512.39	11,963.75	14.476.13	13,160.13
3	13,160.13	1,316.01	13,160.12	14.476.13	0
$N + 1$	0				

We recognize that the interest payments decrease and the repayment amounts increase. The repayment amounts T_k follow a geometric series with the factor q .

(A.3) Due to the renovation of his house the house owner needs to take on a loan with an amount of 36,000 EUR which should be repaid within the next three years. The interest rate equals 10% and the annuity payments should be made at the end of each year. Please draw a repayment plan.

$$(2): D_{k-1} = D_0 * \frac{(q^N - q^{k-1})}{q^N - 1} \Rightarrow D_{2-1} = 36,000 * \frac{1.1^3 - 1.1^{2-1}}{1.1^3 - 1} = 25,123.87 \text{ EUR}$$

$$(3): I_k = r * D_{k-1} \Rightarrow I_2 = r * D_1 = 0.10 * 25,123.87 \text{ EUR} = 2,512.39 \text{ EUR}$$

$$(4): T_k = A - I_k \Rightarrow T_2 = 14,476.13 - 2,512.39 = 11,963.75 \text{ EUR}$$

(A.4) An interest rate of 5% p.a. is applied to a loan of 20,000 EUR. The loan should be repaid by 10 equal annuity payments at the end of each year.

- (a) Please calculate the annuity payment.
- (b) Determine the remaining loan amount after six years.
- (c) Determine the repayment in the sixth year.

$$(a): A = D_0 * q^N * \frac{q - 1}{q^N - 1} = 20,000 * \frac{1.05^{10} * (1.05 - 1)}{1.05^{10} - 1} = 2,590.09 \text{ EUR}$$

$$(b) D_k = D_0 * \frac{q^N - q^k}{q^N - 1} = 20,000 * \frac{1.05^{10} - 1.05^6}{1.05^{10} - 1} = 9,184.34 \text{ EUR}$$

$$(c1) T_1 = A - r * D_0 = 2,590.09 - 0.05 * 20,000 = 1,590.09 \text{ EUR}$$

$$(c2) T_6 = q^{k-1} * T_1 = 1.05^5 * 1,590.09 = 2,029.40 \text{ EUR}$$

Annuity repayment

k	$SD_k = D_{k-1}$	I_k	T_k	A_k	D_k
1	20,000	1,000	1,590	2,590	18,410
2	18,410	920	1,670	2,590	16,740
3	16,740	837	1,753	2,590	14,987
4	14,987	749	1,841	2,590	13,147
5	13,147	657	1,933	2,590	11,214
6	11,214	561	2,029	2,590	9,184
7	9,184	459	2,131	2,590	7,053
8	7,053	353	2,237	2,590	4,816
9	4,816	241	2,349	2,590	2,467
10	2,467	123	2,467	2,590	0
$N + 1$	0				

★ **(A.5)** A loan should be repaid by 15 annual payments of 8,000 EUR at the end of each year. The interest rate equals 5%. How much can the initial loan amount at most be?

1

2

3

$$A = 75,012.13 \text{ EUR}$$

$$B = 83,037.25 \text{ EUR}$$

$$C = 90,340.19 \text{ EUR}$$

$$D = 101,426.26 \text{ EUR}$$

$$(1): A = D_0 * \frac{q^N * (q - 1)}{q^N - 1} \Leftrightarrow D_0 = A * \frac{q^N - 1}{q^N * (q - 1)}$$

$$(2): D_0 = 8,000 * \frac{1.05^{15} - 1}{1.05^{15} * (1.05 - 1)} = 83,037.25 \text{ EUR}$$

(A.6) A loan of 75,000 EUR is repaid by annual payments of 9,000 EUR. Interest rates are 8%.

(a) Please determine how long it takes to pay back the loan.

$$(1): A = D_0 * \frac{q^N * (q - 1)}{q^N - 1}$$

$$(2): D_0 * q^N * (q - 1) - A * q^N = -A$$

$$(3): q^N = \frac{A}{A - D_0 * (q - 1)} \Leftrightarrow q^N = \frac{1}{1 - \frac{D_0 * (q - 1)}{A}}$$

$$(4): N = \frac{-\ln\left(1 - \frac{D_0 * (q - 1)}{A}\right)}{\ln(q)} = \frac{-\ln\left(1 - \frac{75,000 * (1.08 - 1)}{9,000}\right)}{\ln(1.08)}$$

$$(5): N = 14.27 \text{ years}$$

(A.6) A loan of 75,000 EUR is repaid by annual payments of 9,000 EUR. Interest rates are 8%.

(b) Please determine the remaining loan amount at the last full year.

$$(b1): D_{14} = D_0 * q^{14} - A * \frac{q^{14} - 1}{q - 1}$$

$$(b2): D_{14} = 75,000 * 1.08^{14} - 9,000 * \frac{1.08^{14} - 1}{1.08 - 1} = 2,355.24 \text{ EUR}$$

(A.6) A loan of 75,000 EUR is repaid by annual payments of 9,000 EUR. Interest rates are 8%.

(c) What is the final annuity amount that needs to be paid in the final period?

$$(c): A_{15} = D_{14} * q = 2,355.24 * 1.08 = 2,543.66 \text{ EUR}$$

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In some cases it makes sense to start repaying the loan after some payment-free time periods

Two cases need to be distinguished

1. No payments within the payment-free time
 - During the payment-free time the loan amount increases by the interest claims.
2. Interest payments within the payment-free time
 - The loan amount remains constant, however the interest claims need to be paid. The presented formulas from the repayment calculations can be applied.

(A.7) 6% interest rate needs to be paid yearly on a loan of 70,000 EUR. After five payment-free years the loan should be repaid with 10 equal annuity payments. Please calculate the annuity.

(a) Assume that interests are paid within the payment-free time.

(b) Assume that no interests are paid within the payment-free time.

(a1): *Interests are paid: Initial loan amount remains unchanged*

$$(a2): A = D_0 * \frac{q^N * (q - 1)}{q^N - 1} = 70,000 * \frac{1.06^{10} * (1.06 - 1)}{1.06^{10} - 1} = 9,510.76 \text{ EUR}$$

(b1): *Interest are not paid: Initial loan amount increases by unpaid interest claims*

$$(b2): D'_0 = 70,000 * 1.06^5 = 93,675.79 \text{ EUR}$$

$$(b3): A = D'_0 * \frac{q^N * (q - 1)}{q^N - 1} = 93,675.79 * \frac{1.06^{10} * (1.06 - 1)}{1.06^{10} - 1} = 12,727.54 \text{ EUR}$$

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(A.8) A second-semester student assumes that he needs to finance the student fees, the costs of accommodation, and the living expenses by a student loan. The student plans to finish his studies right in time after 8 semesters. While he is looking for student loan offers, he finds two contracts with different conditions. Both contracts will pay him an annuity of 300 EUR at the end of each month starting in his third semester.

(a) Please calculate the total loan amount if the student does not pay any interest payments during his years of study. Please assume an monthly interest rate of 0.5%.

(a1): *Calculation of the future value of the annuity*

$$(a2): FV = C * \frac{q^N - 1}{q - 1} = 300 * \frac{1.005^{12*3} - 1}{1.005 - 1} = 11,800.83 \text{ EUR}$$

★ (A.8) A second-semester student assumes that he needs to finance the student fees, the costs of accommodation, and the living expenses by a student loan. The student plans to finish his studies right in time after 8 semesters. While he is looking for student loan offers, he finds two contracts with different conditions. Both contracts will pay him an annuity of 300 EUR at the end of each month.

For the redemption of his loan the student is offered two repayment plans. The first option is **income-independent** and the student will pay 2,300 EUR as an annuity for the next six years at the end of each year to repay the loan. The second option will offer him an **income-based repayment plan**. The student will pay 4.5% of his yearly income within the first 8 years as an employee. How much does the student need to pay back if he assumes an annual income of (i) 40,000 EUR, (ii) 45,000 EUR, and (iii) 50,000 EUR?

(b) Which option should the student choose under the given income assumptions if interest rates are 5%?

$A = \text{Income} - \text{Independent Plan}$

1

2

3

$B = \text{Income} - \text{Dependent Plan}$

(b1): *Income – independent option*

$$(b1): D_0 = A * \frac{q^N - 1}{q^N * (q - 1)} = 2,300 * \frac{1.05^6 - 1}{1.05^6 * (1.05 - 1)} = 11,674.09 \text{ EUR}$$

(b2): *Income – dependent option*

$$(b2): D_0 = A * \frac{q^N - 1}{q^N * (q - 1)}$$

$$(b2i): D_0 = 0.045 * 40,000 * \frac{1.05^8 - 1}{1.05^8 * (1.05 - 1)} = 11,633.78 \text{ EUR}$$

$$(b2ii): D_0 = 0.045 * 45,000 * \frac{1.05^8 - 1}{1.05^8 * (1.05 - 1)} = 13,088.01 \text{ EUR}$$

$$(b2iii): D_0 = 0.045 * 50,000 * \frac{1.05^8 - 1}{1.05^8 * (1.05 - 1)} = 14,542.23 \text{ EUR}$$

$$(b2iv): 11,674.09 = 0.045 * \text{Income}_{BE} * \frac{1.05^8 - 1}{1.05^8 * (1.05 - 1)} \Leftrightarrow \text{Income}_{BE} = 40,138.59 \text{ EUR}$$